

What if AI could read every pathology report and support medical decisions and prognosis?

From pathology reports to structured clinical insight
at scale.

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- Imagine processing every pathology report a hospital has ever generated.
- We build a DT that reads, extracts, and cross-references at massive scale.
- AI does not replace clinical rules it enhances data analysis.

State of the Art: Latest High-Impact Papers

Top Papers (2024–2026)

- 1 **Eminaga et al.** (2024) *Sci. Reports* (Nature)
“Critical Evaluation of AI as a Digital Twin of Pathologists for Prostate Cancer Pathology” : vPatho system
- 2 **Marchal** (2025) *Nat. Biotechnology*
“Applications of Digital Twins in Medicine” : Perspective on DT frameworks
- 3 **Görtz et al.** (2026) *Nat. Rev. Urology*
“Digital Twins for Personalized Treatment in Uro-Oncology in the Era of AI” : Review

💡 Their Novelty

- ✓ **vPatho:** First systematic benchmark of AI as a “digital twin” of a pathologist; compared with 6 human experts on Gleason grading
- ✓ **Marchal:** Conceptual framework for bidirectional patient–model data flow in clinical DTs
- ✓ **Görtz:** Comprehensive DT roadmap for uro-oncology integrating imaging, genomics & treatment simulation

✕ Their Limitations

- Single cancer type (prostate) focus
- Image-only analysis : no NLP (No expert feedback) on pathology *reports*
- No cross-referencing against AJCC knowledge base
- No automated data discovery or crawling pipeline

- Most AI pathology systems operate in isolation from clinical standards.
- Eminaga (vPatho): first AI-vs-pathologist benchmark on Gleason grading.
- Marchal: bidirectional patient–model data flow framework.
- Görtz: DT roadmap for uro-oncology with imaging and genomics.
- Common gap: image-only, single cancer, no NLP on reports, no AJCC.
- We propose a clinically grounded DT that processes reports at scale.

How We Differ: Our DT Health Pathology Approach

Key Differentiators

Feature	State of Art	Our Approach
Data Input	Images only	Reports + Clinical
Analysis	AI grading	Cross-Ref AJCC KB
Staging	AI predicts	Rule-based (fixed)
Cancer Scope	Single type	Multi-cancer
Data Discovery	Static	MCP Crawl Agent
Human Loop	None	mRAG HILF
LLM Stack	N/A	KG

Disruptive Outcomes

- NLP-first Digital Twin for pathology text reports
- Staging follows **AJCC fixed rules** AI enhances, extracts & cross-references data against the AJCC knowledge base
- MCP-powered crawling agent for automated real-time data discovery and ingestion
- Scenario-based Knowledge Graph with LLM orchestration

Core Insight

The AI pipeline does **not replace** rule-based AJCC staging. Instead, LLMs extract structured fields from free-text reports and cross-reference them against the AJCC book, enabling richer analysis and expert validation.

- We process free-text reports and clinical records, not images.
- AJCC staging is deterministic AI extracts and cross-references, not predicts.
- MCP crawling agent automates data discovery in real time.
- Human-in-the-loop via mRAG keeps clinicians in control.
- System works across multiple cancer types, not just prostate.

Metrics, Hypothesis & Data Architecture

Research Hypothesis

"A Digital Twin architecture that uses LLMs to extract, harmonize, and cross-reference pathology report data against the AJCC knowledge base with human-in-the-loop feedback can further analyse multimodal clinical data at scale and potentially support medical prognosis, enabling richer insight discovery across multiple cancer types by processing massive volumes of pathology reports and clinical records that would be infeasible for manual expert review."

Evaluation Metrics

- **Extraction Accuracy:** Precision, Recall, F1 for entity/field extraction from reports
- **Cross-Ref. Consistency:** Agreement rate between LLM output and AJCC rule-based staging
- **Inter-rater Agreement:** Cohen's κ between DT output and experts
- **Data Quality:** QA/QC pass rate, completeness, provenance
- **Efficiency:** Expert review time reduction (%), cycles saved
- **Statistical Validation:** Mann–Kendall test, variable importance

Data Sources & Pipeline

- 📖 **AJCC Cancer Staging Manual** Full book as TNM knowledge base
- 📄 **Pathology Reports** Free-text (breast cancer primary, extensible)
- 🏥 **Clinical Data** Demographics, treatment, lab results
- 🤖 **MCP Crawling Agent** Automated data discovery & ingestion
- ⚙️ **Data Warehouse** Harmonized repository + provenance tracking

Technology Stack

- Data:** Pathology Reports + AJCC + Clinical + MCP Crawl
- Models:** GCN, LSTM, Knowledge Graph, mRAG
- LLM Orchestration:** Task Planning + Executing LLMs
- Deployment:** Docker / Kubernetes / On-Premise

- Hypothesis: LLM-based DT can analyse multimodal data at scale and support prognosis.
- We use the entire AJCC book as a structured knowledge base.
- Primary data: free-text pathology reports, starting with breast cancer.
- Metrics: extraction F1, AJCC cross-ref consistency, Cohen's kappa, QA/QC rates.
- Efficiency measured by expert review time reduction.
- Statistical validation via Mann–Kendall and variable importance.

TCGA via GDC: Open Dataset & Research Hypothesis

What TCGA/GDC Provides

- **Pathology Reports:** Free-text surgical/pathology PDFs for 33 cancer types (~11,000 cases)
- **Structured TNM:** AJCC stage, T/N/M fields, histology, grade per case
- **Clinical Outcomes:** Overall survival, disease-free survival, treatment records
- **Multi-omics:** Genomics, proteomics, imaging extensible for future fusion
- **Open Access:** Programmatic REST API + gdc-client Python library; no credentials for open-tier data

Programmatic Access

```
POST https://api.gdc.cancer.gov/files
{
  "filters": {
    "op": "and", "content": [
      { "op": "=", "content": {
        "field": "cases.project_id",
        "value": "TCGA-BRCA" } },
      { "op": "=", "content": {
        "field": "data_type",
        "value": "Pathology_Report" } }
```

TCGA-Grounded Hypothesis

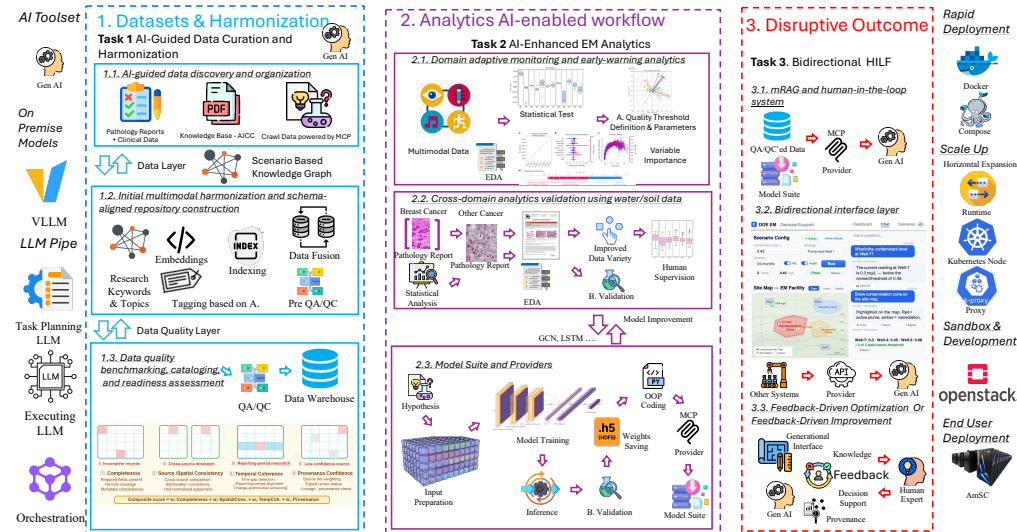
*"By applying the DT Health Pathology NLP pipeline to **TCGA open-access pathology reports**, we hypothesise that LLM-extracted TNM fields will achieve $F1 \geq 0.85$ against the structured AJCC ground truth labels already present in GDC, and that the resulting cross-referenced knowledge graph will reveal **statistically significant associations** between report-derived biomarkers and survival outcomes across ≥ 3 cancer types validating the pipeline at scale before deployment on proprietary institutional data."*

Why TCGA is Ideal for Validation

- Ground-truth TNM labels → direct F1 / AJCC consistency measurement
- Free-text reports + structured outcomes → end-to-end prognosis pipeline test
- Multi-cancer scope → tests generalization before clinical deployment
- No PHI barrier for open tier → fast iteration and reproducibility

- TCGA/GDC is the ideal open benchmark: free-text reports AND ground-truth TNM labels in the same dataset.
- We can compute F1 for NLP extraction directly against GDC structured fields.
- Survival data (OS, DFS) lets us test whether extracted biomarkers are prognostically informative.
- No credentials needed for open-tier; MCP crawling agent can hit the REST API directly.
- Starting with TCGA-BRCA (breast) then extending to LUAD, COAD, etc.
- This slide bridges the hypothesis (Slide 3) to the architecture (Slide 4).

Architecture Overview



- Data layer: reports, clinical records, AJCC KB ingested via MCP agent.
- Model layer: GCN, LSTM, knowledge graph, multimodal RAG.
- LLM orchestration coordinates task planning and execution.
- Human-in-the-loop interface for expert review and feedback.
- Containerised with Docker/Kubernetes, deployable on-premise.

Goal

Build a **Digital Twin (DT)** that ingests free-text clinical pathology reports, extracts structured oncological data using LLMs, and cross-references it against the AJCC Cancer Staging knowledge base enabling **automated, scalable prognosis support** across multiple cancer types with human-in-the-loop validation.

Key Constraint

AI does not predict staging.
AJCC TNM staging follows deterministic, rule-based criteria. The LLM *extracts* and *cross-references* data it **never overrides** clinical rules.

What the System Does

- 1 **Ingest** free-text pathology reports (TCGA/GDC + institutional)
- 2 **Extract** TNM fields, histology & biomarkers via LLMs
- 3 **Cross-reference** against AJCC fixed staging rules
- 4 **Enrich** with a multi-cancer Knowledge Graph
- 5 **Surface** insights via mRAG + expert feedback loop

- The core goal: DT that reads free-text reports and cross-references against AJCC at scale.
- Key constraint is critical for clinical credibility: AI extracts and cross-references, never predicts staging.
- AJCC rules are deterministic LLM output is validated against them, not used as a substitute.
- Five-step pipeline: ingest, extract, cross-ref, enrich KG, surface to clinicians.
- Starting with TCGA-BRCA (breast cancer), extensible to 33 cancer types.

Research Hypotheses

Three Testable Hypotheses

H₁ Extraction Quality
An LLM-based NLP pipeline applied to TCGA open-access pathology reports will extract TNM staging fields with **F1 $\geq$ 0.85** against GDC ground-truth labels.

H₂ Prognostic Value
The cross-referenced Knowledge Graph will identify **statistically significant associations** (Mann–Kendall, $p < 0.05$) between report-derived biomarkers and patient survival outcomes across ≥ 3 cancer types.

H₃ Human-in-the-Loop Benefit
Expert feedback will improve inter-rater agreement (Cohen’s κ) between DT output and pathologist consensus by ≥ 10 pp after one HILF feedback cycle.

How Each is Measured

- **H₁**: Precision, Recall, F1 on NLP entity extraction vs. GDC structured TNM fields
- **H₂**: Mann–Kendall test + variable importance on KG-derived biomarker–survival pairs
- **H₃**: Cohen’s κ before vs. after one expert review cycle

Validation Dataset

Primary: TCGA-BRCA (breast cancer)
Secondary: TCGA-LUAD, TCGA-COAD

- Privacy: de-identification before any data reaches LLMs.
- Report variability handled by multimodal harmonization layer.
- Hallucination mitigated by AJCC rule-based guardrails.
- Expert bottleneck addressed with confidence scoring and selective review.
- Four phases: data curation, AI analytics, HILF, scale-up.